

1. a) Approximately 50% of the human genome is repetitive.

b) SINEs, which are short interspersed nuclear elements, occur with the largest number of copies at 1,797,575. However, LINEs, long interspersed nuclear elements, cover the largest fraction of the genome at 21%.

c) One class of transposable elements are retrotransposons, which are transcribed into RNA intermediates and then reverse-transcribed into cDNA for genome integration. This is "copy-and-paste". Another class are DNA transposons, which use a DNA intermediate to directly "cut-and-paste" or "peel-and-paste" DNA from one location to another. LINEs and SINEs are both the first class, retrotransposons.

d) Both numbers were off, but the amount of the genome that is comprised of LINEs was more off. They assumed only 5% of the genome consisted of LINEs, when in reality about 21% does. This is a difference of about 16%.

2. a) A single-strand of DNA has two main properties that allow it to be a "template". DNA consists of 4 unique bases, A, C, G, T, that make up a molecular code since they participate in complementary base pairing. Each base only pairs with one other (A-T, C-G), and this relationship is symmetrical, meaning whole DNA strands are complementary to each other. With only one strand, it is possible to reconstruct the complementary strand by matching each base on the single strand with its singular complement. In addition, DNA has directionality, so there is a 5' and 3' end. This means that given only one strand, it is possible to reconstruct the entire sequence and direction of the second strand. Since it is possible to determine both DNA strands from a single one, it can be called a template. Cells make use of this property to duplicate DNA to replicate. Proteins cannot be templates because they lack the two properties that DNA has. The amino acids that make up a protein sequence do not have complements, and the codons that make them up are not unique. For example, the amino acid leucine has 6 different RNA anticodons that pair with it, so the DNA sequence that can produce an amino acid sequence isn't unique. Thus, it cannot be determined what the exact DNA sequence is from the protein sequence, which is why proteins cannot be called templates.

b) The terminator ensures that during each cycle of the sequencer, only one base is added. In actual cells, the DNA synthesis reaction happens extremely quickly, so without a terminator, an unknown number of bases could be added to the growing strand every cycle. Sequencing by synthesis depends on clusters of identical DNA strands growing together, so the variable base additions would also throw clusters out of sync, preventing the correct base for every position from being recognized. Since the goal of sequencing is to identify each and every base of the sequence, controlling the reaction to prevent both unintended additions and unsynchronized growth of the sequence are vital.

c) One reason that the cluster is necessary is that an individual fluorescent molecule is not bright enough on its own. When excited, the signal would not be sufficiently intense to detect and call the base. Having a whole cluster present means there will be many molecules fluorescing in the same area, making it easier to detect and call the base of. One problem introduced by using a cluster is that errors during cycles can result in missed or extra bases being added, which can result in a blurred signal. As

more mistakes occur, individual strands in a cluster start to fluoresce different colors, making it difficult to tell what base is present at each position in the sequence. With individual molecules, this ambiguity in base-calling would not be present, though the mistakes might.

3. a) There are 16 length-4 palindromes. We can choose the first half of the length to be any letter, and the other two have to complement them.

b) A length-3 palindrome isn't possible because it would require the 2nd base to be a complement of itself, and this is not true for any of the possible bases. To generate a DNA palindrome, we need to enforce that the first half of the sequence complements the second half symmetrically, as if they can pair with each other if they were folded in the middle. This works with even lengths but not with odd lengths since no base can pair with itself.

4. a) $\frac{1}{3}$ of the 192 codon mutations result in a different amino acid. 3 codons result in a different amino acid for any of the other 3 third bases, 26 result in a different amino acid for 2 possible third bases, and 3 results in a different amino acid for 1 different third base. This results in 64 codons that differ, and $\frac{64}{192} = \frac{1}{3}$.

b) In the standard code, the RNA codon UGA is a stop codon, but in human mitochondria, this is not the case. Instead of being a stop codon in human mitochondria, UGA encodes tryptophan (W) instead. Krüger A, Kovalchuk D, Shiriaev D, Rorbach J. Decoding the Enigma: Translation Termination in Human Mitochondria. Hum Mol Genet 2024;33(R1):R42-R46. doi:10.1093/hmg/ddae032.

5. a) The naive algorithm tries $m \cdot n + 1$ alignments. Since $m = 18$ and $n = 4$, 15 total alignments are tried. 22 character comparisons are made.

b) The smallest number of character comparisons that native matching can make is $1 \text{ character} \cdot 15 \text{ alignments} = 15 \text{ character comparisons}$. An example of this would be $T = \text{AAAAAAAAAAAAAAAAAAAA}$ and $P = \text{GGGG}$. Since there are no G's in T, every alignment would stop after just the first letter of P was compared, leading to 15 total alignments. On the other hand, the largest number of comparisons is $4 \text{ characters} \cdot 15 \text{ alignments} = 60 \text{ character comparisons}$. An example of this would be $T = \text{AAAAAAAAAAAAAAAAAAAA}$ and $P = \text{AAAA}$. Since every character matches, for every one of the 15 possible alignments all 4 characters in P are compared. This results in a total of 60 comparisons.

c) We can immediately rule out alignments 7 and 8 because we know the $T[7]$ and $T[8]$ would not match with $P[0]$. This is a usage of the good suffix rule since the only prefix matching the suffix is 'A', requiring two skips. Thus, from alignment 6 we know immediately to skip to alignment 9.